



## Technology Data Sheet (TDS)

### TINMORRY PC-GF

#### Version 1.0

#### • Basic Introduction

TINMORRY PC-GF is a glass fiber reinforced PC composite. It retains the toughness, temperature resistance, and durability of conventional PC, and has significantly improved anti-warping, relatively higher strength, stiffness, layer strength, surface hardness, and wear resistance. It also gives the prints' surface a special texture and feel brought by glass fiber, and hides the layer lines, giving it a delicate and consistent appearance. In addition, it can be printed at very high speed (300 mm/s) on most printers on the market, and has a rich variety of colors to choose from, making it an ideal fiber-reinforced engineering filaments.

Note: 1. Since the PC-GF filament contains glass fiber, please use a nozzle with a diameter of  $\geq 0.4$  mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them. 2. Please use an enclosed printer to print it to reduce warping and ensure high enough layer strength.

#### • Specifications

| Subjects            | Data                                                              |
|---------------------|-------------------------------------------------------------------|
| Diameter            | $1.75 \pm 0.03$ mm                                                |
| Net filament weight | $1000\text{ g} \pm 0.5\%$                                         |
| Length              | $330 \pm 5$ m                                                     |
| Spool material      | PC ( (Temperature resistance: $120\text{ }^{\circ}\text{C}$ )     |
| Spool size          | External Diameter: 198 mm; Inner Diameter: 56 mm;<br>Width: 65 mm |

## • Recommended Printing Settings

| Subjects                          | Data                                                                                                                                                    |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drying settings before printing   | 85 ± 5 °C, 6 - 8 h<br>(Blast drying oven, or filament drying box)                                                                                       |
| Printing temperature and humidity | ≤ 75 °C, ≤ 20% RH<br>( Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set the temp to be about 70 °C ) |
| Storage temperature and humidity  | ≤ 30 °C, ≤ 20% RH ( Sealed with desiccant )                                                                                                             |
| Compatible support material       | Itself or support filament for PC                                                                                                                       |
| Compatible printer type           | enclosed-frame (recommended)<br>Open-frame (not recommended)                                                                                            |
| Compatible nozzle material        | Hardened steel                                                                                                                                          |
| Compatible nozzle size            | 0.4, 0.6 (recommended), 0.8 mm                                                                                                                          |
| Compatible plate type             | Textured PEI Plate or other plate                                                                                                                       |
| Plate surface preparation         | Glue                                                                                                                                                    |
| Nozzle temperature                | 270 - 300 °C                                                                                                                                            |
| Bed temperature                   | 90 - 110 °C<br>(depending on the printer and plate types)                                                                                               |
| Printing speed*                   | Max: 300 mm/s                                                                                                                                           |
| Chamber temperature               | < 80 °C                                                                                                                                                 |

\*When printed with Bambu X1C and P1S, and the nozzle temperature is 285 °C, the max printing speed can reach 300 mm/s.

## • Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PC-GF have been tested and are shown in the tables below.

### • Physical Properties

| Subjects                        | Testing Methods    | Data                   |
|---------------------------------|--------------------|------------------------|
| Density                         | ISO 1183           | 1.26 g/cm <sup>3</sup> |
| Saturated water absorption rate | 25 °C , 55% RH     | 0.23%                  |
| Melt index                      | 230 °C, 2.16 kg    | 18.2 ± 2.1 g/ 10 min   |
| Glass transition temperature    | DSC, 10 °C/min     | 116 °C                 |
| Crystallization temperature     | DSC, 10 °C/min     | N / A                  |
| Melting temperature             | DSC, 10 °C/min     | 224 °C                 |
| Vicar softening temperature     | ISO 306, GB/T 1633 | 111 °C                 |
| Heat deflection temperature 1   | ISO 75, 1.82 MPa   | 98 °C                  |
| Heat deflection temperature 2   | ISO 75, 0.45 MPa   | 107 °C                 |

### • Mechanical Properties

| Properties                    | Testing Methods    | Data                             |
|-------------------------------|--------------------|----------------------------------|
| Tensile strength (XY)         | ISO 527, GB/T 1040 | $61 \pm 2$ MPa                   |
| Tensile strength (Z)          | ISO 527, GB/T 1040 | $46 \pm 3$ MPa                   |
| Young's modulus (XY)          | ISO 527, GB/T 1040 | $2752 \pm 168$ MPa               |
| Young's modulus (Z)           | ISO 527, GB/T 1040 | $2046 \pm 158$ MPa               |
| Breaking elongation rate (XY) | ISO 527, GB/T 1040 | $3.2\% \pm 0.8\%$                |
| Breaking elongation rate (Z)  | ISO 527, GB/T 1040 | $1.4\% \pm 0.7\%$                |
| Bending strength (XY)         | ISO 178, GB/T 9341 | $101 \pm 3$ MPa                  |
| Bending strength (Z)          | ISO 178, GB/T 9341 | $73 \pm 3$ MPa                   |
| Bending modulus (XY)          | ISO 178, GB/T 9341 | $2869 \pm 174$ MPa               |
| Bending modulus (Z)           | ISO 178, GB/T 9341 | $2118 \pm 153$ MPa               |
| Impact strength (XY)          | ISO 179, GB/T 1043 | $38.4 \pm 2.1$ kJ/m <sup>2</sup> |
| Impact strength (Z)           | ISO 179, GB/T 1043 | $17.3 \pm 2.2$ kJ/m <sup>2</sup> |

### • Chemical and Other Properties

| Subjects                      | Data                                                              |
|-------------------------------|-------------------------------------------------------------------|
| Component                     | PC, glass fiber                                                   |
| Skin irritation               | May cause allergy in some people, and wearing gloves is suggested |
| Resistance to water           | Yes                                                               |
| Resistance to oil and grease  | Resistant to most kinds of daily oil and grease                   |
| Resistance to organic solvent | Not resistant to some organic solvents                            |
| Resistance to weak acid       | Yes                                                               |
| Resistance to strong acid     | Not resistant                                                     |
| Resistance to weak alkali     | Yes                                                               |
| Resistance to strong alkali   | Not resistant                                                     |
| Flammability                  | Yes                                                               |
| Odor during printing          | Odorless                                                          |
| Odor during combustion        | Pungent                                                           |

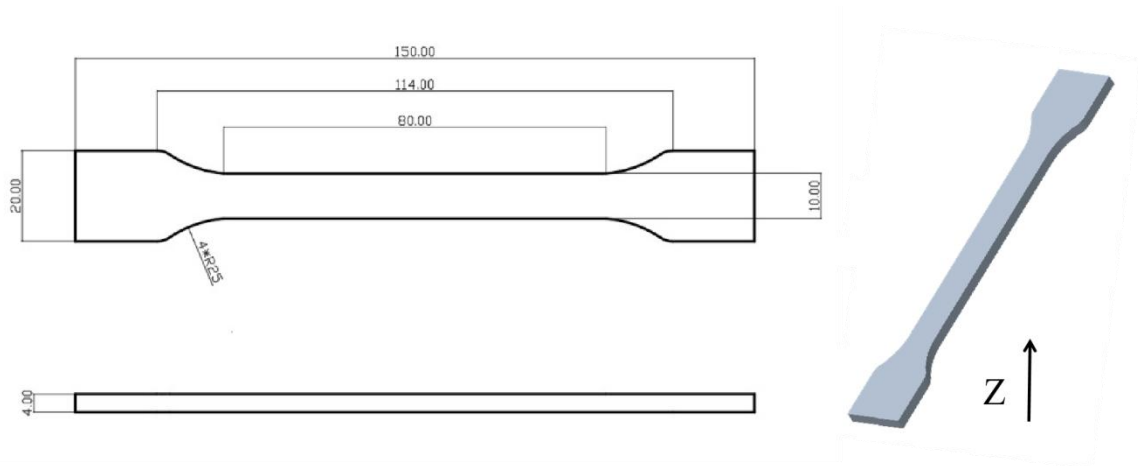
### • Specimen Info

| Subjects           | Data                            |
|--------------------|---------------------------------|
| Size               | Shown at the following pictures |
| Nozzle temperature | 285 °C                          |
| Bed temperature    | 100 °C                          |
| Printing speed     | XY: 150 mm/s; Z: 80 mm/s        |
| Infill density     | 100%                            |
| Infill pattern     | Concentric                      |

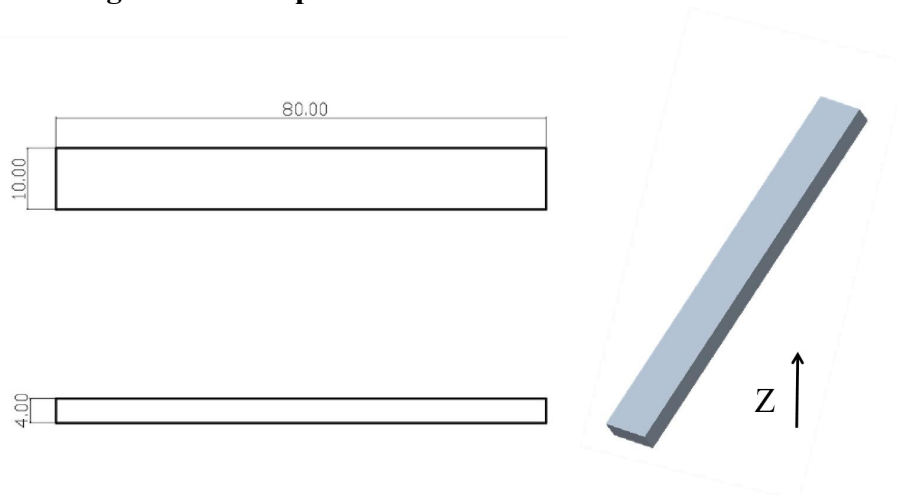
Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties of TINMORRY PC-GF prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.

## 1. Tensile Test



## 2. Bending Test and Impact Test



## • **Disclaimer**

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.